

CSCI-665 Foundations of Algorithms

Homework 5

due Thursday, November 14, 2013

Reading Chapters 24, 25

1. Run the Bellman-Ford algorithm on the directed graph on Figure 24.4 page 652, using vertex t as the source. In each pass, relax edges in the same order as in the figure, and show the d and π values after each pass.
2. Run the Floyd-Warshall algorithm on a subgraph of the graph of Figure 25.2 page 691 induced by 4 vertices $\{1, 2, 4, 5\}$. Show the matrix $D^{(k)}$ that results for each iteration of the outer loop.
3. Solve exercise 24.1-3, page 654.
4. Solve exercise 24.1-6, page 655.
5. Solve exercise 24.3-6, page 663.
6. Solve exercise 24.4-1, page 669.
7. Solve exercise 25.2-6, page 700.